

Quiz 6

1

FILL IN THE BLANK

0.25 / 0.25



Sensor fusion involves combining data from multiple sensors to obtain a more accurate and reliable perception of the environment.

2

MULTIPLE CHOICE

0 / 0.25



In a typical autonomous navigation pipeline, which controller type is commonly used for smooth motion execution?

☐ A. Model Predictive Control (MPC)

☒ B. Open-loop control

☐ C. bang-bang control

☐ D. Proportional Control (P)

3

FILL IN THE BLANK

0.25 / 0.25



Tactile sensors detect physical contact or touch, allowing robots to interact with objects in their environment.

4

MULTIPLE CHOICE

0.25 / 0.25



In the context of robot perception, what does SLAM stand for?

☐ A. Simultaneous Location and Mapping

☐ B. Synchronized Localization and Monitoring

☐ C. Sensor Location and Measurement

- ☒ D. Simultaneous Localization and Mapping

5 FILL IN THE BLANK

0.25 / 0.25



Environmental sensors are used to measure environmental conditions such as temperature and humidity.

6 TRUE/FALSE

0.25 / 0.25



In robot perception, object recognition primarily relies on deep learning and feature matching techniques.

☒ True

☐ False

7 MULTIPLE CHOICE

0.25 / 0.25



What is the main function of LiDAR sensors in robotics?

☐ A. To detect sound

☒ B. To measure distances and create 3D maps

☐ C. To sense temperature changes

☐ D. To capture images

8 MULTIPLE CHOICE

0.25 / 0.25



Which statement best describes when Dijkstra's algorithm is preferred over A*?

☒ A. Dijkstra's is optimal for graphs with nonnegative weights and no informative heuristic

☐ B. Dijkstra's uses a heuristic to estimate the remaining cost

☐ C. A* and Dijkstra have identical behavior on all grids

☐ D. A* is always faster than Dijkstra in all graphs

9 FILL IN THE BLANK

0.25 / 0.25



Force sensors detect force and torque applied to an object, which is crucial in tasks like robotic surgery and assembly.

10 MULTIPLE CHOICE

0.25 / 0.25



RRT is particularly suited for high-dimensional and dynamically changing spaces.

☐ A. False

☒ B. True

11 MULTIPLE CHOICE

0.25 / 0.25



Which of the following is a challenge in robot perception related to the environment?

☒ A. Changing lighting and weather conditions

☐ B. Battery life

☐ C. Real-time data processing

☐

D. Sensor durability

12

MULTIPLE CHOICE

0.25 / 0.25



Which type of proprioceptive sensor is used to measure rotational velocity?

☒ A. Gyroscope

☐ B. Force Sensor

☐ C. Accelerometer

☐ D. Encoder

13

MULTIPLE CHOICE

0.25 / 0.25



A* guarantees the shortest path even if the heuristic is poorly chosen or inconsistent.

☒ A. False

☐ B. True

14

MULTIPLE CHOICE

0.25 / 0.25



SLAM requires prior knowledge of the map to begin localization.

☐ A. True

☒ B. False

15

MULTIPLE CHOICE

0.25 / 0.25



What is the primary purpose of robot perception?

- ☒ A. To interpret data from sensors to understand the environment
- ☐ B. To power the robot
- ☐ C. To communicate with other robots
- ☐ D. To perform maintenance tasks

16

TRUE/FALSE

0.25 / 0.25



Kalman Filter is a technique used in sensor fusion to combine predictions from a model with sensor data.

- ☒ True
- ☐ False

17

MULTIPLE CHOICE

0.25 / 0.25



What is the primary role of perception in autonomous navigation?

- ☒ A. Sensing and interpreting the environment
- ☐ B. Localizing the robot within a map
- ☐ C. Maintaining a stable control loop
- ☐ D. Executing the planned trajectory

18

MULTIPLE CHOICE

0.25 / 0.25



Sensors such as cameras and LiDAR can be fused to improve perception accuracy.

☒ A. True☐ B. False

19

TRUE/FALSE

0.25 / 0.25



Infrared sensors can be used for night vision and obstacle detection.

☒ True☐ False

20

MULTIPLE CHOICE

0.25 / 0.25



Which of the following is a type of proprioceptive sensor?

☒ A. Encoder☐ B. Microphone☐ C. LiDAR☐ D. Camera